

Mohammed Abdul Omer

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PROFESSIONAL SUMMARY

Computer Science student (graduating 2026) with hands-on experience building production-ready generative AI systems and NLP solutions. Demonstrated expertise in developing LLM-powered applications, RAG architectures, and scalable API services. Proven track record of automating workflows and delivering intelligent systems that enhance user experiences and operational efficiency. Seeking AI/ML engineering internship opportunities.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, C++
AI/ML Frameworks: LangChain, OpenAI API, Google Gemini, TensorFlow, PyTorch, scikit-learn
Backend & APIs: FastAPI, REST APIs, Flask, Node.js
Data & Databases: Vector Databases (Pinecone, Chroma), PostgreSQL, MongoDB, Pandas, NumPy
GenAI & NLP: RAG Pipelines, Prompt Engineering, LLM Fine-tuning, Embedding Models, Semantic Search
Tools & Platforms: Git, Docker, Gradio, Jupyter, Google Cloud Platform, AWS, Linux

PROFESSIONAL EXPERIENCE

Artificial Intelligence Intern <i>TechZone Software Academy for Training and Research</i>	March 2025 – June 2025 <i>Hyderabad, India</i>
- Architected and deployed 4+ production-grade generative AI applications serving 500+ users, including an AI-powered doubt-solving tutor delivering contextual explanations and real-time academic support using LLMs and advanced prompt engineering - Engineered an automated PTE mock-test platform with AI-driven evaluation capabilities for listening, reading, and writing assessments, improving student test readiness by 40% through personalized feedback mechanisms - Implemented Retrieval-Augmented Generation (RAG) pipelines with vector databases (Pinecone/Chroma) to enhance response accuracy by 35% and reduce hallucinations in LLM outputs through semantic search and context injection - Developed scalable microservices using FastAPI and REST architecture, integrating OpenAI APIs and LangChain workflows with structured JSON pipelines for reliable AI service orchestration - Built intelligent automation systems including quiz generation engines and attendance tracking modules, reducing manual workload by 60% across academic departments through workflow optimization - Optimized LLM performance through systematic prompt engineering, achieving 25% reduction in API costs while maintaining output quality through iterative testing and model evaluation strategies	

KEY PROJECTS

AI Educational Assistant Platform <i>Python, FastAPI, OpenAI API, LangChain, RAG, Vector DB</i>	March 2025 – June 2025
- End-to-end AI platform integrating doubt resolution, automated test generation, and intelligent feedback systems - Leveraged RAG architecture to retrieve relevant educational content and generate contextually accurate responses - Deployed RESTful microservices handling concurrent API requests with <200ms latency for real-time interactions	

EDUCATION

Lords Institute of Engineering & Technology <i>Bachelor of Engineering in Computer Science</i>	Expected May 2026 <i>Hyderabad, India</i>
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KEY ACHIEVEMENTS

- Delivered 4 production-ready AI applications reducing manual academic operations by 60% organization-wide
- Improved LLM response accuracy by 35% through RAG implementation and vector database optimization
- Reduced API costs by 25% through strategic prompt engineering and model selection strategies
- Enabled 500+ students to access personalized AI-driven learning and automated assessment tools